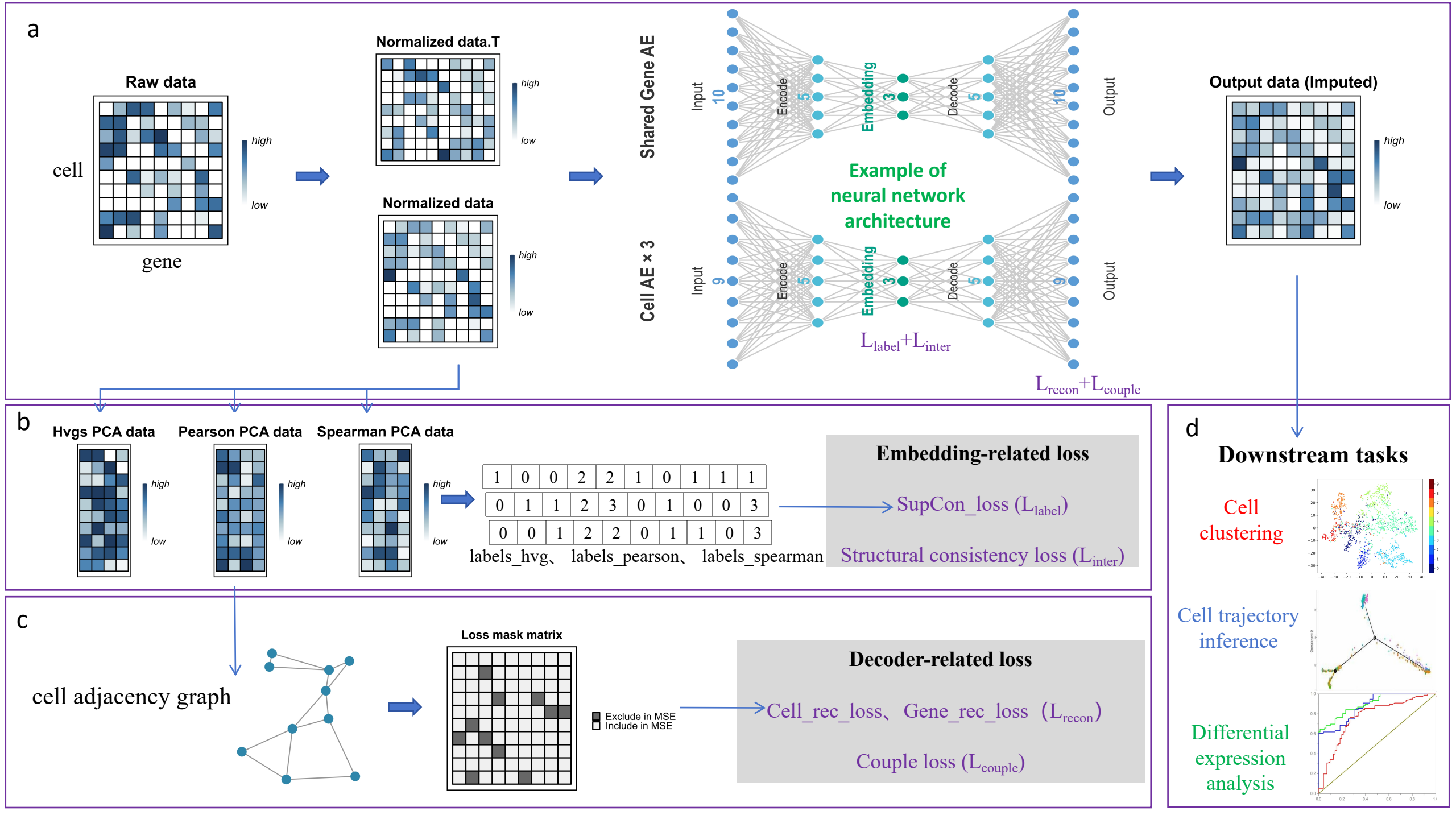

a
Raw data
cell
gene
high
low
Normalized data.T
Normalized data
Shared Gene AE
Cell AE × 3
Input
Encode
Embedding
Decode
Output
Example of neural network architecture
$L_{label}+L_{inter}$
$L_{recon}+L_{couple}$
Output data (Imputed)
b
Hvgs PCA data
Pearson PCA data
Spearman PCA data
labels_hvg、 labels_pearson、 labels_spearman
Embedding-related loss
SupCon_loss ($L_{label}$)
Structural consistency loss ($L_{inter}$)
c
cell adjacency graph
Loss mask matrix
Exclude in MSE
Include in MSE
Decoder-related loss
Cell_rec_loss、Gene_rec_loss ($L_{recon}$)
Couple loss ($L_{couple}$)
d
Downstream tasks
Cell clustering
Cell trajectory inference
Differential expression analysis